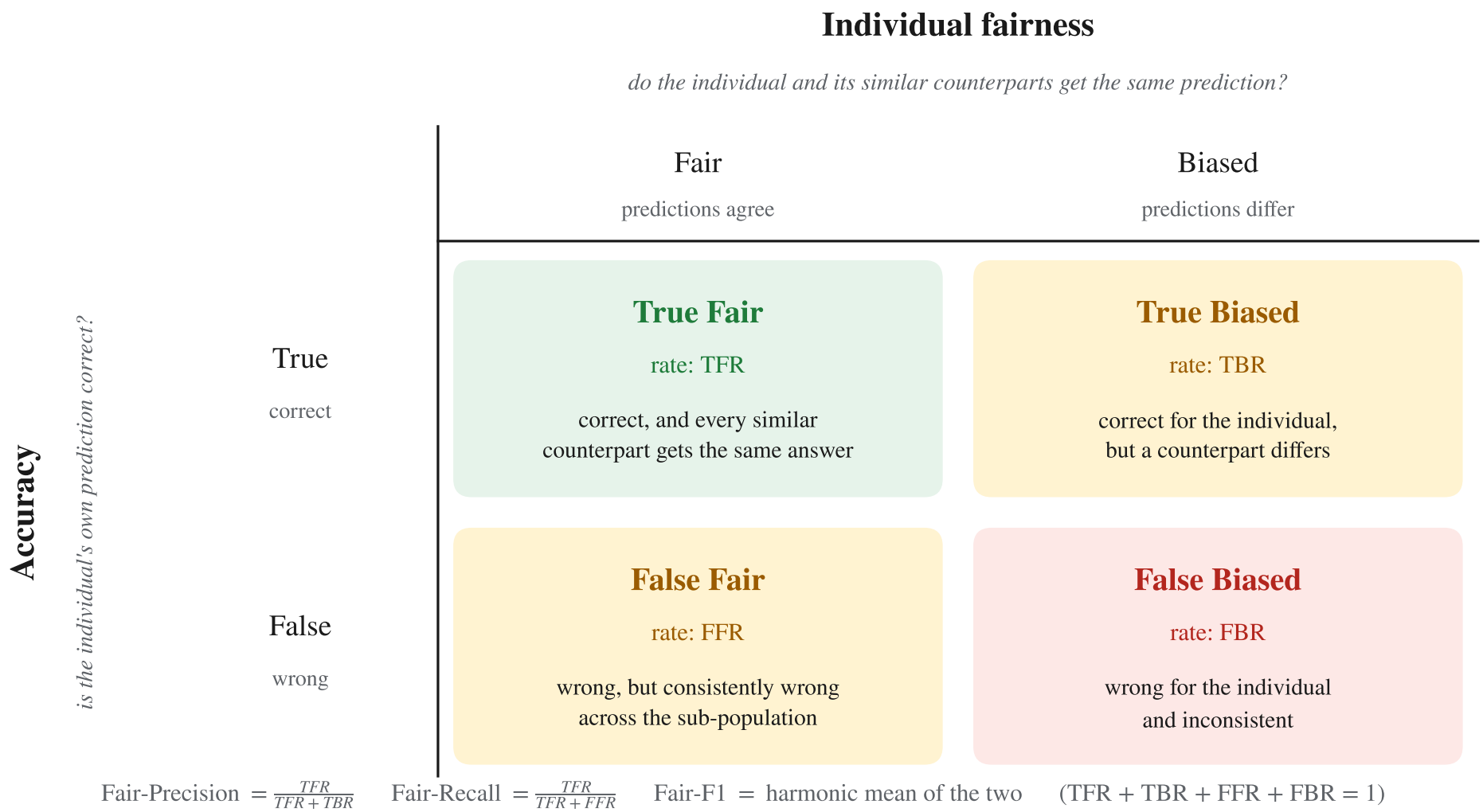
Individual fairness
do the individual and its similar counterparts get the same prediction?
Fair
predictions agree
Biased
predictions differ
Accuracy
is the individual's own prediction correct?
True
correct
False
wrong
True Fair
rate: TFR
correct, and every similar counterpart gets the same answer
True Biased
rate: TBR
correct for the individual, but a counterpart differs
False Fair
rate: FFR
wrong, but consistently wrong across the sub-population
False Biased
rate: FBR
wrong for the individual and inconsistent
Fair-Precision = TFR / (TFR + TBR)
Fair-Recall = TFR / (TFR + FFR)
Fair-F1 = harmonic mean of the two
(TFR + TBR + FFR + FBR = 1)